StrykerJS - Mutation Testing

Version: 10.0.0 **Website:** stryker-mutator.io

The Problem

Your tests pass — but do they actually catch bugs? A test suite can hit 100% line coverage while still missing entire classes of logic errors. **Mutation testing** answers the question your coverage report can't: *if the code were broken, would your tests notice?*

What Stryker Does

Stryker introduces small, deliberate bugs into your code called **mutants** (e.g. `>` → `<=`, `+` → `-`, `true` → `false`). It runs your test suite against each one. If your tests fail, the mutant is killed ✅ — your tests caught it. If they pass, the mutant **survived** ❌ that bug would ship undetected.

The result is a **mutation score** (killed ÷ total mutants). Unlike coverage, you can't raise it without writing assertions that genuinely constrain behavior.

Setup

Prerequisites: Node v22+, and npm, yarn, or pnpm.

```
# Run the interactive setup wizard from your project root
npx stryker init

# In a monorepo, run it inside the specific package instead
cd packages/your-package
npm init stryker@latest
```

Answer the prompts:

Prompt	Recommended answer
Framework preset	None/other (unless using Angular/CRA/Vue)
Test runner	jest, vitest, jasmine — match your project
Build command	Leave empty unless you need a compile step
Reporters	`html`, `clear-text`, `progress` (defaults)
Package manager	Match your project (npm / yarn / pnpm)
Config file type	JSON

Edit `stryker.config.json` – the key settings.

```
{
  "$schema": "./node_modules/@stryker-mutator/core/schema/stryker-schema.json",
  "testRunner": "vitest",
  "coverageAnalysis": "perTest",
  "mutate": [
    "src/**/*.ts",
    "!src/**/*.test.ts",
    "!src/**/*.spec.ts"
  ],
  "reporters": ["html", "clear-text", "progress"],
  "thresholds": { "break": 60 }
}
```

> `coverageAnalysis`: "perTest" is the biggest speed win — it runs only the tests that cover each mutant instead of the whole suite every time.

Add report output to `.gitignore`:

```
reports/mutation
```

Running Stryker

```
# Scope to one file while iterating (much faster)
npx stryker run --mutate "src/utils/myFile.ts"

# Full run over everything in your mutate config
npx stryker run

# CI: only mutate files changed vs. base branch
npx stryker run --since main

# Open the HTML report
open reports/mutation/mutation.html    # Mac
start reports/mutation/mutation.html   # Windows
```

The HTML report overlays every mutant on your source, **green = killed, red = survived.** Click any red mutant to see the exact change your tests missed, then write a test to kill it.

When to Use It ✅/ When Not To ❌

✅ Good fit	❌ Not a good fit
Logic-heavy, unit-testable code	Visual / UI testing
Verifying test quality before a release	Tight deadlines (full runs are slow)
Finding gaps in an existing test suite	Code with no unit tests yet
JS, TS, C# (Stryker.NET), or Scala (Stryker4s)	End-to-end / integration test suites

Cost: Completely free. No account required.

Tutorial Video

https://www.youtube.com/watch?v=clGgBeSfm_8